

Turing Instabilities - Exploring Morphogenesis

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1 Introduction

Morphogenesis is defined as the biological process through which cells, tissues and organisms acquire dimension and form [1]. As a result, the system that exhibits morphogenesis, although originally homogeneous, may later develop a pattern or structure due to an instability of the homogeneous equilibrium triggered by random disturbances[2]. While studying the changes of state of a developing organism one should take into account,

- the changes of position and velocity
- the stresses as given by elasticities and motion
- the chemical reactions
- the diffusion of the chemical substances.

The counterintuitive claim proposed was that, in the presence of diffusion, spatial patterns could form in chemical systems that are otherwise stable[2]. These patterns are called Turing patterns and are caused by diffusion-driven instability, aka Turing instability[3]. Our aim is to explore mathematically a chemical system that exhibits Turing patterns under diffusion. Typical chemical systems describe the reaction and diffusion between two chemical substances.

Consider two chemical substances U and V that can react with each other[4]. The reaction system will be of the form

$$\begin{cases} \frac{du}{dt} = au + bv \\ \frac{dv}{dt} = cu + dv \end{cases} \quad (1)$$

with real coefficients a, b, c, d denoting effective reaction rate parameters. Here, $u(t)$ & $v(t)$ are concentrations of U & V at time $t \geq 0$. For this ODE system, $(u, v) = (0, 0)$ is a steady state. It is asymptotically stable if all eigenvalues of the coefficient matrix A have negative real parts. This leads to the following conditions on the trace and determinant of the coefficient matrix, required for the stability of the steady state.

$$tr(A) = a + d < 0 \quad \text{and} \quad det(A) = ad - bc > 0 \quad (2)$$

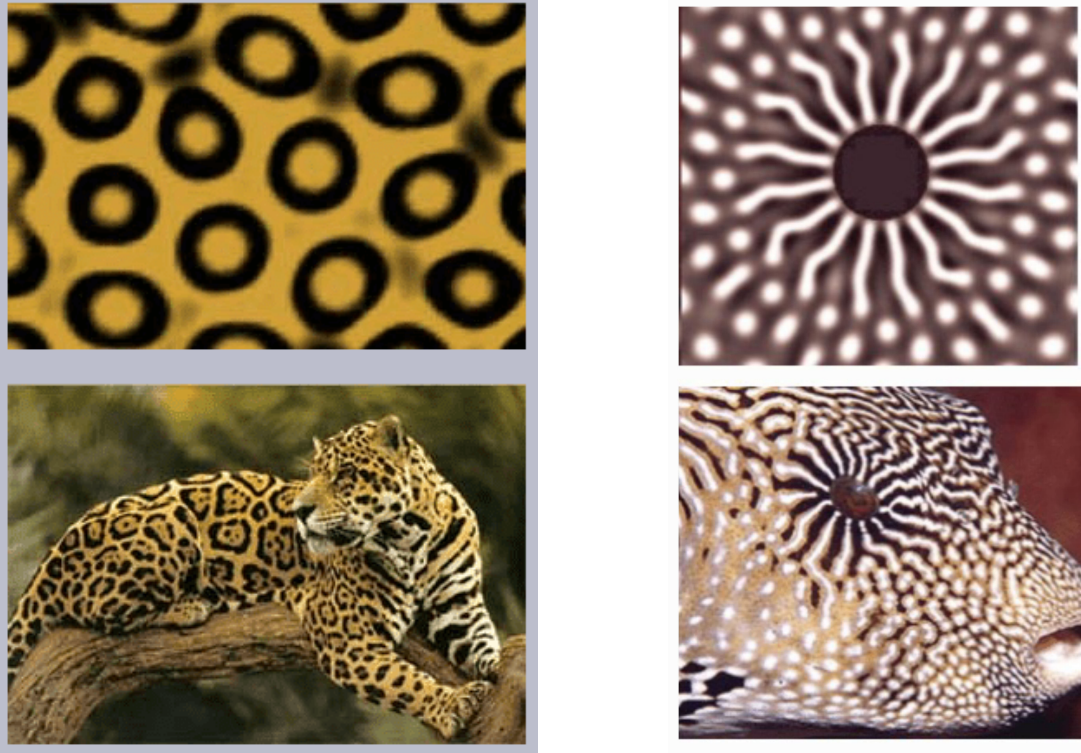
where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

In order to add diffusion to the system, the following coefficients σ_u & σ_v , which denote the diffusion rates for u & v , respectively, and $\nabla^2 u$ & $\nabla^2 v$, which account for

the spatial diffusion of u & v , respectively, are added to the ODE. Let $\Omega \subset \mathbb{R}^2$ be a bounded domain; then the reaction diffusion system can be defined as:

$$\begin{cases} \frac{\partial u}{\partial t} - \sigma_u \nabla^2 u = au + bv \\ \frac{\partial v}{\partial t} - \sigma_v \nabla^2 v = cu + dv \end{cases} \quad \forall (x, t) \in \Omega \times (0, \infty) \quad (3)$$

with Dirichlet boundary conditions. Hence, the steady state of the ODE system (1), $(u, v) = (0, 0)$, is also a steady state of the PDE system.



(a) A Turing pattern at the top and a real jaguar at the bottom.[5]

(b) Patterns formed around the eye of a fish.[6]

Figure 1: Turing patterns in nature.

2 Stability Analysis

Turing Instability Theorem. *Let $\Omega \subset \mathbb{R}^2$ be a bounded domain. Consider the reaction-diffusion system*

$$\begin{cases} \frac{\partial u}{\partial t} - \sigma_u \nabla^2 u = au + bv \\ \frac{\partial v}{\partial t} - \sigma_v \nabla^2 v = cu + dv \end{cases} \quad \forall (x, t) \in \Omega \times (0, \infty)$$

subject to Dirichlet boundary conditions. Assume the conditions (2) hold with $a > 0, d < 0$. Fix $\sigma_v > 0$. Then when, σ_u is sufficiently small the steady state $(u, v) = (0, 0)$ is linearly unstable. Moreover, only finitely many eigenmodes are unstable [4].

Proof. Because $a > 0, d < 0$, species u acts as an activator and v as an inhibitor. Let $L = -(\nabla)^2$ be the Laplace operator with Dirichlet boundary conditions which is positive, self-adjoint, and its inverse L^{-1} is compact. By the spectral theorem, there exists a complete orthonormal basis $\{w_k\}_{k \geq 1}$ of eigenfunctions and an associated, strictly increasing sequence of positive eigenvalues $0 < \lambda_1 \leq \lambda_2 \leq \dots$, with

$$-(\nabla)^2 w_k = \lambda_k w_k, \quad \lambda_k \xrightarrow{k \rightarrow \infty} \infty.$$

In order to project the reaction - diffusion system onto these eigen functions, the time-dependent functions u and v can be decomposed as follows

$$\begin{aligned} u(t) &= \sum_{k=1}^{\infty} \alpha_k(t) \mathbf{w}_k, \\ v(t) &= \sum_{k=1}^{\infty} \beta_k(t) \mathbf{w}_k. \end{aligned}$$

Where,

$$\begin{aligned} \alpha_k(t) &= \langle u(t, \cdot), \mathbf{w}_k \rangle, \\ \beta_k(t) &= \langle v(t, \cdot), \mathbf{w}_k \rangle. \end{aligned}$$

The orthonormal property of $\mathbf{w}_k$ guarantees that,

$$\langle w_j, w_k \rangle = \delta_{jk} = \begin{cases} 1; j = k \\ 0; j \neq k \end{cases}$$

After taking the scalar product of each equation in the reaction-diffusion system with a single eigenfunction $\mathbf{w}_k$ it can be simplified to the following.

$$\begin{cases} \frac{d\alpha_k(t)}{dt} + \sigma_u \lambda_k \alpha_k(t) = a\alpha_k(t) + b\beta_k(t), \\ \frac{d\beta_k(t)}{dt} + \sigma_v \lambda_k \beta_k(t) = c\alpha_k(t) + d\beta_k(t). \end{cases} \quad (4)$$

Instability occurs when at least one small perturbation eventually grows large over time. The system (4) can be written as

$$\begin{bmatrix} \dot{\alpha}_k(t) \\ \dot{\beta}_k(t) \end{bmatrix} = \begin{bmatrix} -\sigma_u \lambda_k + a & b \\ c & -\sigma_v \lambda_k + d \end{bmatrix} \begin{bmatrix} \alpha_k \\ \beta_k \end{bmatrix} \quad (5)$$

The solutions to (4) can be expressed as follows.

$$\alpha_k(t) = e^{\lambda t} \bar{\alpha}_k, \quad \beta_k(t) = e^{\lambda t} \bar{\beta}_k$$

where λ are the eigenvalues of the coefficient matrix in (5) and $\bar{\alpha}_k, \bar{\beta}_k(t)$ are time dependent constants.

Here λ can be referred to as the growth function and for instability to occur $Re(\lambda)$ must be positive, which is necessary and sufficient. Otherwise, $e^{\lambda t}$ will remain bounded, and so the corresponding linear combinations as a result the equilibrium will remain stable. Therefore, for instability, we need $Re(\lambda) > 0$. Substitute for α_k and β_k in the system (4)

$$\begin{cases} \lambda \bar{\alpha}_k + \sigma_u \lambda_k \bar{\alpha}_k = a \bar{\alpha}_k + b \bar{\beta}_k, \\ \lambda \bar{\beta}_k + \sigma_v \lambda_k \bar{\beta}_k = c \bar{\alpha}_k + d \bar{\beta}_k. \end{cases} \quad (6)$$

For the 2×2 linear system in (6) a non-trivial solution for $\bar{\alpha}_k$ and $\bar{\beta}_k$ exists iff,

$$\det \begin{pmatrix} \lambda + \sigma_u \lambda_k - a & -b \\ -c & \lambda + \sigma_v \lambda_k - d \end{pmatrix} = 0 \quad (7)$$

Upon expanding the determinant (7), the dispersion relation is obtained:

$$\lambda^2 + \lambda[(\sigma_u + \sigma_v)\lambda_k - \mathcal{T}] + \sigma_u \sigma_v (\lambda_k)^2 - \lambda_k(d\sigma_u + a\sigma_v) + \mathcal{D} = 0. \quad (8)$$

Where, $\mathcal{T} = tr(A) = a + d < 0$ and $\mathcal{D} = det(A) = ad - bc > 0$, defined in (2).

Hence, there exists a solution with exponential growth for eigenvalues λ_k , for which there is a root $\lambda > 0$, for the Dispersion Relation (8). The coefficients for the second order and first order terms of the the Dispersion Relation (8) are positive since $\mathcal{T} < 0$, $\lambda > 0$, $\sigma_u > 0$ and $\sigma_v > 0$ by assumption. So a positive root exists if, and only if, the zeroth order term is negative.

$$\therefore \lambda > 0 \Leftrightarrow (\sigma_u \sigma_v (\lambda_k)^2 - \lambda_k(d\sigma_u + a\sigma_v) + \mathcal{D}) < 0.$$

This inequality can be simplified to the following.

$$(\lambda_k)^2 - \lambda_k \left(\frac{d}{\sigma_v} + \frac{a}{\sigma_u} \right) + \frac{\mathcal{D}}{\sigma_u \sigma_v} < 0. \quad (9)$$

Since $\lambda_k > 0$ and $\frac{\mathcal{D}}{\sigma_u \sigma_v} > 0$ the relation (9) holds when $\frac{d}{\sigma_v} + \frac{a}{\sigma_u} > 0$ is sufficiently large along with $\frac{\mathcal{D}}{\sigma_u \sigma_v}$ sufficiently small.

For a fixed set of coefficients (a, b, c, d) , it is difficult to explain instability using σ_u, σ_v alone. To characterize instability, further information about which eigen functions (w_k , Spatial Frequency) grow is required. For that more information about λ_k is needed, without which it is difficult to point out the exact unstable modes corresponding to a small σ_u .

Set,

$$\theta = \frac{\sigma_u}{\sigma_v}.$$

The relationship in (9) can be further simplified as

$$Q(\lambda_k) = (\lambda_k)^2 - \lambda_k \left(\frac{d\theta}{\sigma_u} + \frac{a}{\sigma_u} \right) + \frac{\mathcal{D}\theta}{\sigma_u^2} < 0. \quad (10)$$

Then the roots of the polynomial λ_k belong to

$$\lambda_k \in [\Lambda_-, \Lambda_+]$$

where

$$\Lambda_{\pm} = \frac{1}{2\sigma_v\theta} \left[(d\theta + a) \pm \sqrt{(d\theta + a)^2 - 4\mathcal{D}\theta} \right]. \quad (11)$$

For instability, the discriminant term (Δ) , $(d\theta + a)^2 - 4\mathcal{D}\theta$, must be positive. Then the inequality (10) is satisfied specifically for $\lambda_k \in [\Lambda_-, \Lambda_+]$. Any Laplace eigenvalue in this interval grow and a Turing pattern can emerge. When the discriminant $((d\theta + a)^2 - 4\mathcal{D}\theta) = 0$, $\Lambda_- = \Lambda_+ = \Lambda_*$ and the inequality (10) is never satisfied. However, $Q(\lambda_*) = 0$ corresponds to a marginally stable mode. When the discriminant $((d\theta + a)^2 - 4\mathcal{D}\theta) < 0$, then $Q(\lambda_k) > 0 \quad \forall \lambda_k > 0$. As a result, the homogeneous steady state remains linearly stable. Figure 2 summarizes how the instability changes as discriminant decreases from positive to negative values for constant parameters $a = 2.0, b = 3.0, c = -4.0, d = -3.0$ & $\sigma_v = 1$ and $\sigma_u > 0$.

Simplifying (11) further gives,

$$\Lambda_{\pm} = \frac{d\theta + a}{2\sigma_v\theta} \left[1 \pm \sqrt{1 - \frac{4\mathcal{D}\theta}{(d\theta + a)^2}} \right]. \quad (12)$$

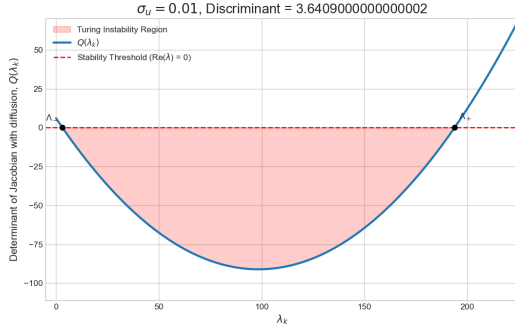
When σ_u is sufficiently small, the regime θ will also be small. Which means that the chemical species u diffuses slowly compared to v . Then, $(d\theta + a) \approx a$. Applying Taylor expansion on (12) gives,

$$\Lambda_{\pm} \approx \frac{a}{2\sigma_v\theta} \left[1 \pm \left(1 - \frac{2\mathcal{D}\theta}{(d\theta + a)^2} \right) \right].$$

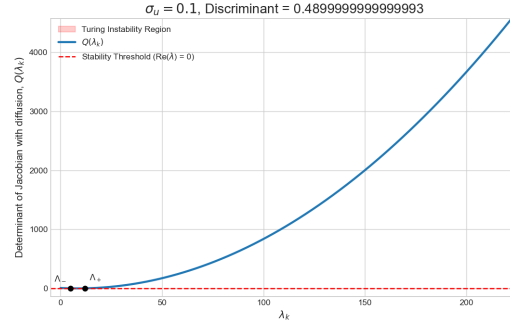
And thus,

$$\Lambda_- \approx \frac{\mathcal{D}}{a\sigma_v} = O(1), \quad \Lambda_+ \approx \frac{a}{\sigma_v\theta} \gg 1.$$

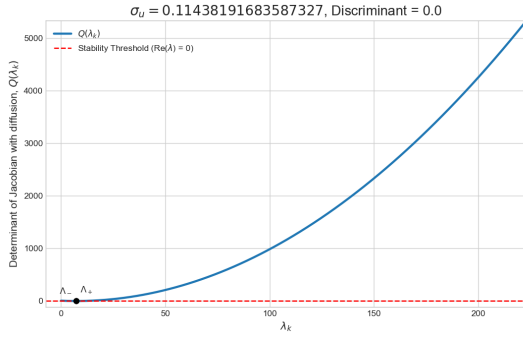
With information on the distribution of λ_k , the unstable modes can be pointed out. Since σ_u is small, that the interval $[\Lambda_-, \Lambda_+]$ is large but at the same time finite. Also, $\lim_{k \rightarrow \infty} \lambda_k = +\infty$. Which means that "not all" but some λ_k will fall in this interval. And there will only be a finite number of unstable modes λ_k . Hence, the homogeneous steady state $(u, v) = (0, 0)$ is linearly unstable when σ_u is small and only a finite number of eigen modes are unstable. $\square$



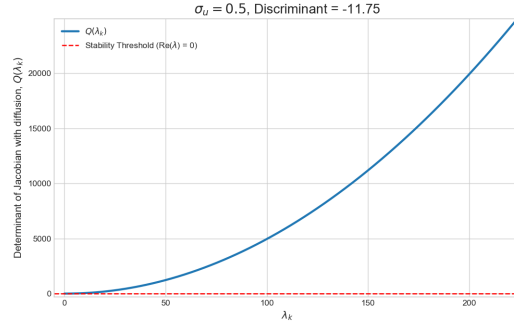
(a) $\Delta > 0$, large unstable region.



(b) $\Delta > 0$, narrow unstable region.



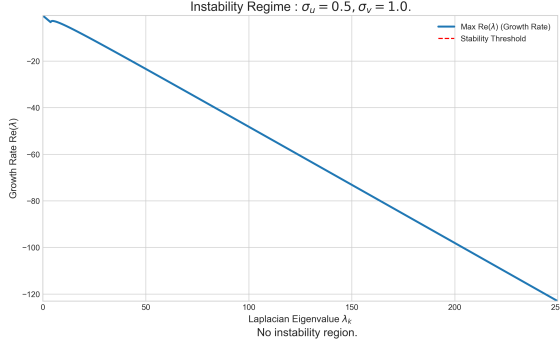
(c) $\Delta = 0$, $\lambda_- = \lambda_+ = \lambda_*$ and $Q(\lambda_*) = 0$. Marginal stability.



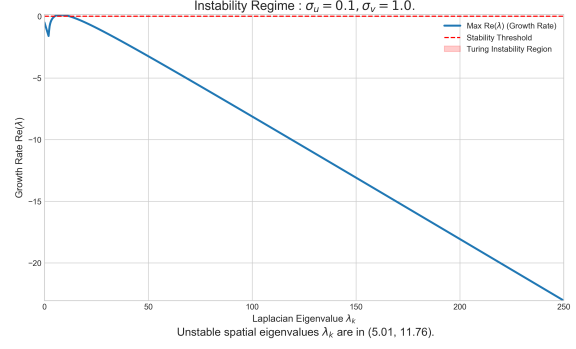
(d) $\Delta < 0$, no unstable modes.

Figure 2: Effect of the discriminant term on the unstable eigen value range.

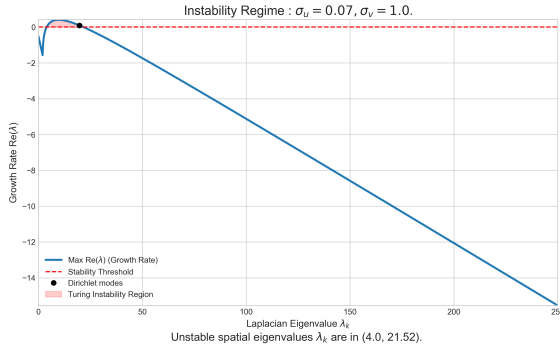
Figure 3 shows the change in the instability regime with changing σ_u . Observe that, as σ_u decrease the instability region is increasing and there is slight or no instability when $\sigma_u \geq 0.1$ for constant parameters $a = 2.0, b = 3.0, c = -4.0, d = -3.0$ & $\sigma_v = 1$. The corresponding unstable Dirichlet eigenvalues that lies in the unstable region is plotted in black dots.



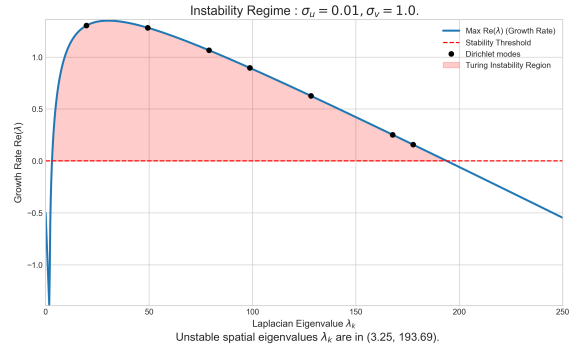
(a) No unstable modes.



(b) Narrow unstable regime, but no Dirichlet unstable modes.



(c) Small unstable regime. Dirichlet unstable eigen values are $\lambda_k = [19.73]$



(d) Large unstable regime. Dirichlet unstable eigen values are $\lambda_k = [19.73, 49.34, 98.69, 167.78, 49.34, 78.95, 128.30, 98.69, 128.30, 177.65, 167.78]$

Figure 3: Effect of σ_u on the instability regime.

3 Numerical Simulation

Consider $\Omega = [0, 1] \times [0, 1] \subset \mathbb{R}^2$ bounded domain,

$$\begin{cases} \frac{\partial u}{\partial t} - \sigma_u \nabla^2 u = au + bv & , x \in \Omega \\ \frac{\partial v}{\partial t} - \sigma_v \nabla^2 v = cu + dv \end{cases}$$

with Dirichlet boundary conditions. Here u and v are the concentrations of the substances U and V respectively, σ_u and σ_v are the diffusion coefficients

[7] A fully implicit finite difference method (FDM) is applied to the system. Time discretization is carried out using the Backward Euler scheme.

Let u^n denote the solution u at time t^n given by $n\Delta t$. The time derivative is approximated as:

$$\frac{\partial u}{\partial t} \approx \frac{u^{n+1} - u^n}{\Delta t}$$

Applying this to the system and evaluating the RHS at the future time step $n + 1$,

$$\begin{aligned}\frac{u^{n+1} - u^n}{\Delta t} &= \sigma_u \nabla^2 u^{n+1} + a u^{n+1} + b v^{n+1} \\ \frac{v^{n+1} - v^n}{\Delta t} &= \sigma_v \nabla^2 v^{n+1} + c u^{n+1} + d v^{n+1}\end{aligned}$$

The domain Ω is discretized with spacing Δx and Δy . Let $u_{i,j}^n$ be the value of u at grid point (x_i, y_i) and time t^n .

The Laplacian $\nabla^2 u$ is approximated using the standard 5-point stencil:

$$\nabla^2 u \approx \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{(\Delta x)^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{(\Delta y)^2}$$

If we assume $\Delta x = \Delta y = h$,

$$\nabla^2 u \approx \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}}{h^2}$$

Similarly for $\nabla^2 v$,

$$\nabla^2 v \approx \frac{v_{i+1,j} + v_{i-1,j} + v_{i,j+1} + v_{i,j-1} - 4v_{i,j}}{h^2}$$

Substituting and rearranging all unknowns in step $n+1$ on the left and knowns from step n on the right,

$$\begin{aligned}(1 - a\Delta t)u_{i,j}^{n+1} - \frac{\sigma_u \Delta t}{h^2}(u_{i+1,j}^{n+1} + u_{i-1,j}^{n+1} + u_{i,j+1}^{n+1} + u_{i,j-1}^{n+1} - 4u_{i,j}^{n+1}) - b\Delta t \cdot v_{i,j}^{n+1} &= u_{i,j}^n \\ (1 - d\Delta t)v_{i,j}^{n+1} - \frac{\sigma_v \Delta t}{h^2}(v_{i+1,j}^{n+1} + v_{i-1,j}^{n+1} + v_{i,j+1}^{n+1} + v_{i,j-1}^{n+1} - 4v_{i,j}^{n+1}) - c\Delta t \cdot u_{i,j}^{n+1} &= v_{i,j}^n\end{aligned}$$

Flattening the 2D grid of values $u_{i,j}^{n+1}$ of size $N \times N$ into a single column vector u^{n+1} of size $N^2 \times 1$. Similarly, we do it for v^{n+1} .

Then the system of equations can be expressed in matrix form as

$$\begin{bmatrix} A_{uu} & A_{uv} \\ A_{vu} & A_{vv} \end{bmatrix} \begin{bmatrix} u^{n+1} \\ v^{n+1} \end{bmatrix} = \begin{bmatrix} u^n \\ v^n \end{bmatrix}$$

where the block matrices are,

- $A_{uu} = (1 - a\Delta t)I - \sigma_u \Delta t \cdot L$
- $A_{uv} = -b\Delta t I$
- $A_{vu} = -c\Delta t I$
- $A_{vv} = (1 - d\Delta t)I - \sigma_v \Delta t \cdot L$

and L is the matrix representing the discrete Laplacian operator which is a sparse matrix of size $N^2 \times N^2$ and I is the identity matrix of size $N^2 \times N^2$

The invertibility of matrix $A = \begin{bmatrix} A_{uu} & A_{uv} \\ A_{vu} & A_{vv} \end{bmatrix}$ can be proven using block matrix analysis. We take the Schur complement of $A_{uu} = A_{vv} - A_{vu}A_{uu}^{-1}A_{uv} =: S$. If the matrix A_{uu} and S are invertible, then the block matrix A is also invertible and A_{uu} and S are invertible when Δt is small. Therefore, we need to make sure Δt is small in our code.

The system of equations are solved using scipy functions like *factorized* and *spsolve*. Following are the results from the numerical implementation of the system.

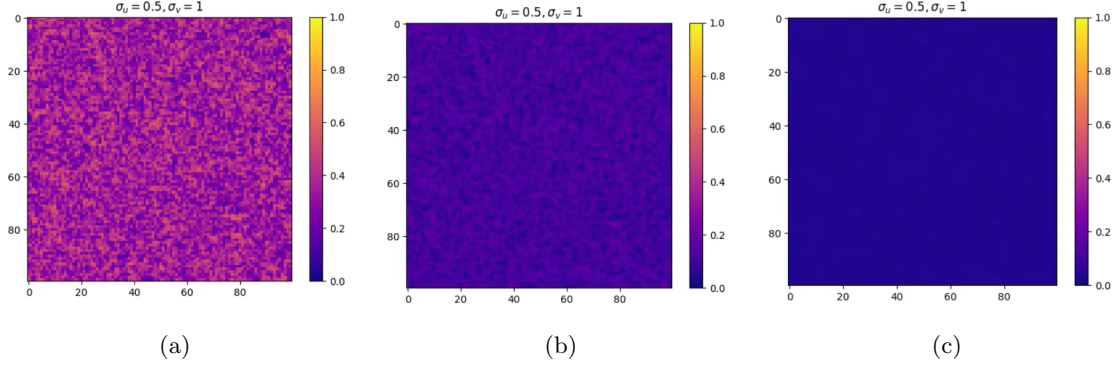


Figure 4: Evolution of the reaction-diffusion system from a perturbed state to stable state when $\sigma_u=0.5$ because it exhibits no instability region as seen in Figure 3(a).

It can also be seen in Figure 3(d) that for $\sigma_u = 0.01$, we have a total of 11(7 distinct) eigenvalues. For $[0, L] \times [0, L]$ with Dirichlet boundary condition, the eigenvalues and eigenfunctions for $k, l \in \mathbb{N}$ are given by

$$\lambda_{kl} = \left(\frac{\pi k}{L}\right)^2 + \left(\frac{\pi l}{L}\right)^2, \quad w_{kl} = \sin\left(\frac{\pi k x}{L}\right) \sin\left(\frac{\pi l y}{L}\right)$$

Taking $L = 1$ and $\lambda_{kl} \approx 19.739$, we have $19.739 = \pi^2(k^2 + l^2) \implies k^2 + l^2 \approx 2$ which gives $k = 1$ and $l = 1$. Therefore, the system is initialized with a sine wave perturbation corresponding to the mode numbers $k = 1$ and $l = 1$ and the results in Figure 5 are observed. Note that the coefficients $a, b, c, d = 2, 3, -4, -3$ remain unchanged.

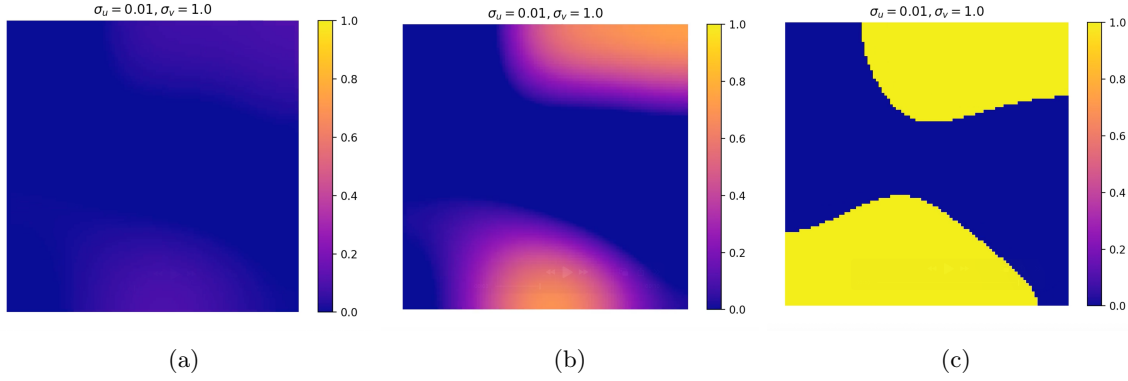


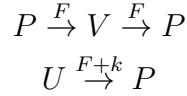
Figure 5: Adding a perturbation to the system in the form of unstable eigenmodes grows, giving the characteristic Turing patterns

4 Application: Gray-Scott Model

The Gray-Scott Model is a reaction-diffusion system modeling the following chemical reaction between two chemical substances U and V [8]:



The reaction consumes V and produces U . To maintain the reaction, the amount of both substances needs to be controlled which is done by adding V at the feed rate F and removing U at the kill rate k . Further, V is produced and destroyed with the same rate F and U decays on it's own with a rate $F + k$,



where P is an inert product and here V diffuses faster than U

The Gray-Scott system is defined by two equations that describe the behaviour of two reacting substances:

$$\begin{aligned} \frac{\partial u}{\partial t} &= u^2 v - (F + k)u + \sigma_u \nabla^2 u \quad \text{for } x \in \Omega, t > 0 \\ \frac{\partial v}{\partial t} &= -u^2 v + F(1 - v) + \sigma_v \nabla^2 v \quad \text{for } x \in \Omega, t > 0 \end{aligned}$$

with periodic boundary conditions

where u and v are the concentrations of the reacting substances and $-(F + k)u$ is the removal term of U and $F(1 - v)$ is the feed term of V

We need to prove Turing instability for this system. First, we need to show stability under the absence of diffusion. Then, we introduce diffusion to the system and check whether a steady point becomes unstable. [9]

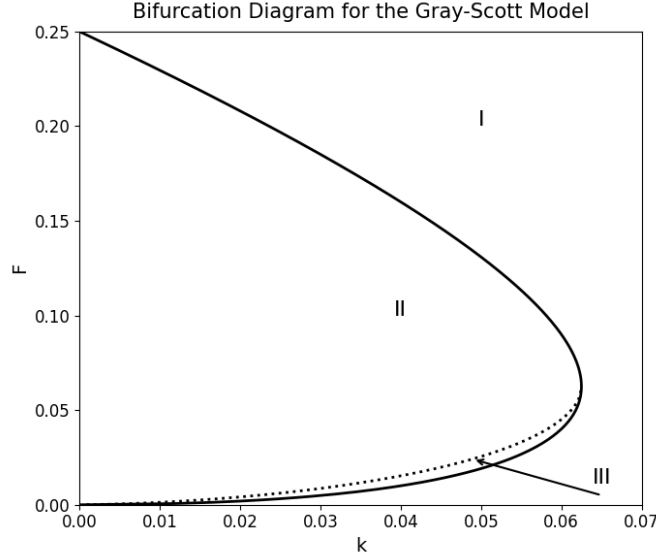


Figure 6: Phase diagram for the steady states (under absence of diffusion). The solid curve indicates the existence of the steady state. In region I only the trivial state exists and inside the solid curve all three states exist. In region II the steady state (U_+, V_+) is linearly stable. Dotted curve denotes the condition for stability of this steady state. Thus, in region III (between the dotted and solid curve) the steady state becomes unstable.

We compute that there are three homogeneous steady states to the Gray-Scott system:

1. The trivial state $(U_0, V_0) = (0, 1)$
2. The non-trivial steady states (U_+, V_+) and (U_-, V_-) which can be expressed as follows,

$$U_{\pm} = \frac{F \pm \sqrt{F^2 - 4(F+k)^2 F}}{2(F+k)}, \quad V_{\pm} = \frac{F \mp \sqrt{F^2 - 4(F+k)^2 F}}{2F}$$

It is clear that the trivial state exists for all parameter values F and k . The non-trivial steady states (U_+, V_+) and (U_-, V_-) exist only when,

$$0 < k < \frac{1}{2}(\sqrt{F} - 2F), \quad 0 < F < \frac{1}{4}$$

which is indicated as the solid curve in Figure 6

It is trivial to test the stability of the three steady states and obtain the following results:

1. The trivial state $(U_0, V_0) = (0, 1)$ is linearly stable
2. The steady state (U_+, V_+) is linearly stable under the condition:

$$F > \frac{1}{2}(-2k + \sqrt{k} - \sqrt{k - 4k^{\frac{3}{2}}})$$

3. The steady state (U_-, V_-) is linearly unstable

We have two steady states which are linearly stable under the absence of diffusion. Now, we consider the system under diffusion and check if the states exhibit Turing instability. For that we need to compute certain conditions called 'Turing conditions' for our system.

$$\begin{aligned}\frac{\partial u}{\partial t} &= f(u, v) + \sigma_u \nabla^2 u \quad \text{for } x \in \Omega, t > 0 \\ \frac{\partial v}{\partial t} &= g(u, v) + \sigma_v \nabla^2 v \quad \text{for } x \in \Omega, t > 0\end{aligned}$$

where $f(u, v) = u^2v - (F + k)u$ and $g(u, v) = -u^2v + F(1 - v)$

The Jacobian of the system can be expressed as,

$$J = \begin{bmatrix} f_u & f_v \\ g_u & g_v \end{bmatrix} = \begin{bmatrix} 2uv - (F + k) & u^2 \\ -2uv & -u^2 - F \end{bmatrix}$$

We use the Turing instability theorem (2) mentioned in the first section to obtain the following 'Turing conditions':

1. $tr(J) = f_u + g_v < 0$
2. $det(J) = f_u g_v - f_v g_u > 0$
3. $\sigma_v f_u + \sigma_u g_v > 0$, $\frac{\sigma_v}{\sigma_u} \neq 1$
4. $(\sigma_v f_u + \sigma_u g_v)^2 - 4\sigma_u \sigma_v det(J) > 0$

Now, to check if the steady states satisfy the Turing conditions,

Consider the trivial state $(U_0, V_0) = (0, 1)$, then the third Turing condition becomes,

$$-\sigma_v(F + k) - \sigma_u F$$

which can never be positive since F, k, σ_u, σ_v are positive constants.

$\implies$ the steady state (U_0, V_0) does not exhibit Turing instability

For the other steady state (U_+, V_+) , to avoid lengthy calculations we perform them numerically in Python to get Figure (7),

From Figure (7) we understand that the steady state (U_+, V_+) exhibits Turing instability in the region T in a certain parameter space (F, k) . We can also infer that, the larger the ratio of the diffusion coefficients, the bigger the Turing region becomes.

Thus, after obtaining all the conditions for Turing instability for the Gray-Scott Model we implement the numerical simulations in Python to generate various patterns.

Turing Instability Regions for the Gray-Scott Model

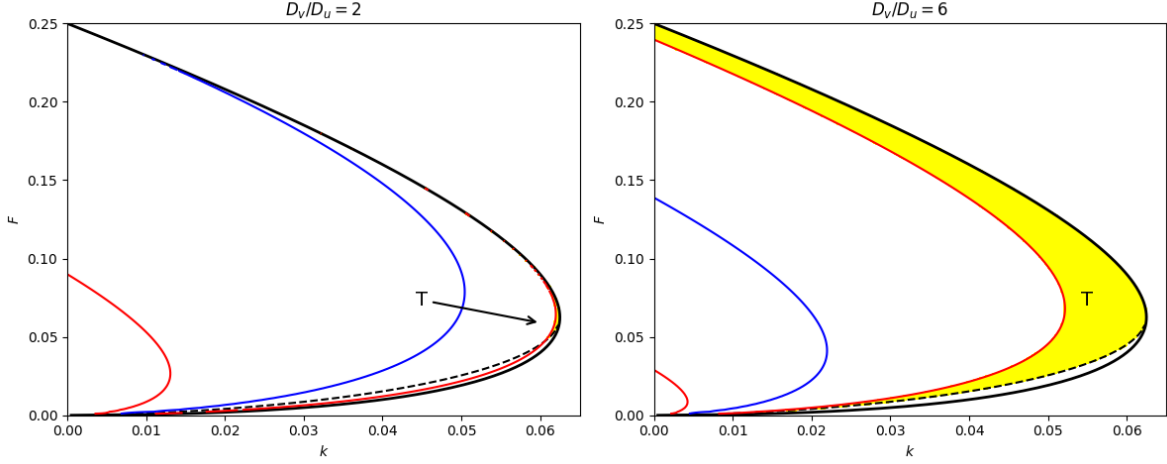


Figure 7: Phase diagram for the steady state (U_+, V_+) (under diffusion). The solid and dashed curves represent the conditions as in the earlier Figure (6). The blue and red curves represent the third and fourth Turing conditions respectively. The region T denotes the Turing instability region.

The implementation in the code is based on the following semi-explicit discretization, Consider the diffusion part of the model:

$$\frac{\partial u}{\partial t} = \sigma_u \nabla^2 u \text{ and } \frac{\partial v}{\partial t} = \sigma_v \nabla^2 v$$

Applying implicit Euler we get,

$$\begin{aligned} \frac{u^* - u^n}{\Delta t} &= \sigma_u \Delta u^* \\ \implies (\mathbb{I} - dt \cdot \sigma_u \cdot \Delta) u^* &= u^n \end{aligned}$$

We do the same for v ,

$$(\mathbb{I} - dt \cdot \sigma_v \cdot \Delta) v^* = v^n$$

where $\mathbb{I}$ is the identity, u^* and v^* denote u and v at time t_{n+1} respectively.

Now, we discretize the space as a 2D grid with $n \times n$ points with unit grid spacing h and approximate the Laplacian with the finite difference method,

$$\begin{aligned} \nabla^2 u &\approx \frac{1}{h^2} (u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}) \\ \nabla^2 v &\approx \frac{1}{h^2} (v_{i+1,j} + v_{i-1,j} + v_{i,j+1} + v_{i,j-1} - 4v_{i,j}) \end{aligned}$$

Flattening U^n and V^n as vectors at time t_n and $\mathcal{L}$ as the discrete Laplacian matrix,

$$(\mathbb{I} - dt \cdot D_u \cdot \mathcal{L}) U^* = U^n$$

$$(\mathbb{I} - dt \cdot D_v \cdot \mathcal{L})V^* = V^n$$

This linear system can be solved with Scipy sparse matrix solver methods like 'spsolve', 'factorized' [10]

We get U^* and V^* as intermediate solutions after diffusion.

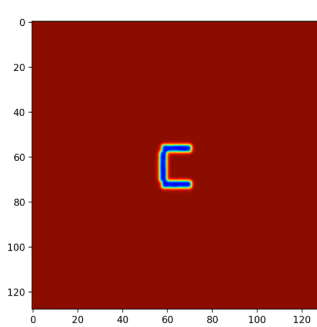
Applying it to the reaction terms explicitly to get u^{n+1} and v^{n+1} ,

$$\begin{aligned} u^{n+1} &= u^* + \Delta t((u^*)^2 v^n - (F + k)u^*) \\ v^{n+1} &= v^* + \Delta t(-(u^{n+1})^2 v^* + F(1 - v^*)) \end{aligned}$$

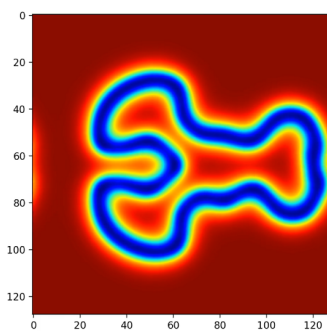
where Δt is the time step and u^* , v^* are the non-flattened matrices.

And so we showcase a number of examples below. Note that all the patterns were generated with time step = 0.1, appropriate values of F and k and a diffusion coefficient ratio $\frac{\sigma_v}{\sigma_u} = 2$.

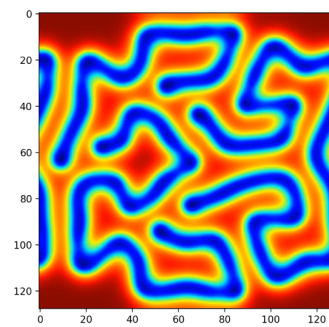
The code will be available on the [Github repository](#)



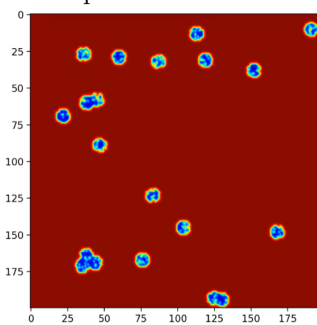
(a) Initial state with 'C' shaped initialization



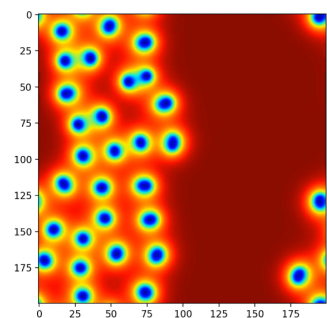
(b) Intermediary state with 'C' shaped initialization



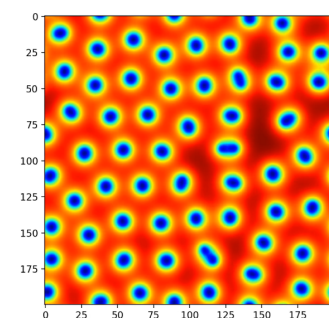
(c) Final state



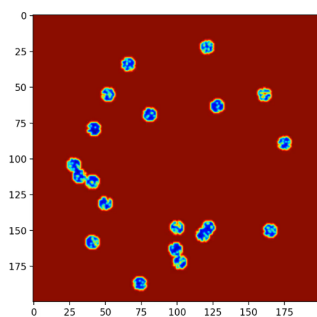
(a) Initial state with random 'blob' shaped initialization



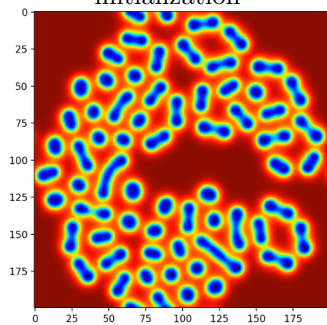
(b) Intermediary state with random 'blob' shaped initialization



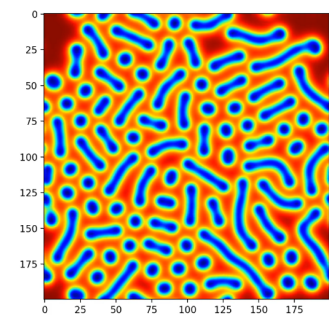
(c) Final state



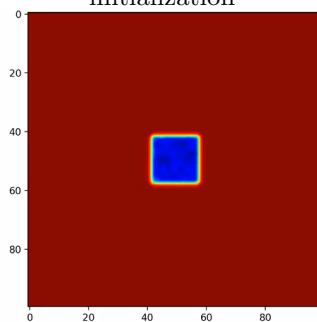
(a) Initial state with random 'blob' shaped initialization



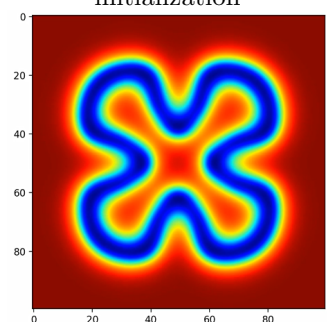
(b) Intermediary state with random 'blob' shaped initialization



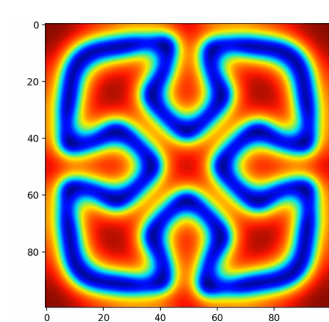
(c) Final state



(a) Initial state with 'square' shaped initialization



(b) Intermediary state with 'square' shaped initialization



(c) Final state

5 References

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